Homework 3
3 February 2025
Total Pages: 8

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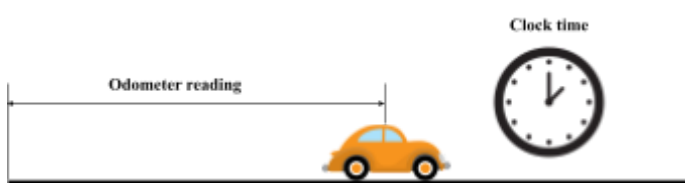
Question 1

Given:
Solar car readings

Clock time (hr:min:sec)	0:00:00	0:59:12	2:01:46	2:58:55	3:47:01	4:13:00	5:36:17
Odometer reading (mi)	102.0	157.8	217.6	264.1	315.2	341.7	420.3
Time interval (hr)							
Distance (mi)							
Average speed (mph)							

Find:
The time interval, distance, average speed, and overall average speed.

Diagram:



Theory:

Finite difference $\Delta x_i = x_{(i+1)} - x_i$

Average Speed $|\bar{v}| = \frac{\Delta x}{\Delta t}$

Assumptions:
To use finite difference, we must assume that our data is sufficiently smooth and differentiable across the given domain.

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Solution:

Clock time (hr:min:sec)	0:00:00	0:59:12	2:01:46	2:58:55	3:47:01	4:13:00	5:36:17
Odometer reading (mi)	102.0	157.8	217.6	264.1	315.2	341.7	420.3
Time interval (hr)	0.99	1.04	0.95	0.80	0.43	1.39	
Distance (mi)	55.8	59.8	46.5	51.1	26.5	78.6	
Average speed (mph)	56.55	57.35	48.82	63.74	61.19	56.63	

First time interval calculation

$$\Delta t_0 = t_1 - t_0 = (0{:}59{:}12) - (0{:}00{:}00) = 0{:}59{:}12 = 0.99\,hr$$

First distance interval calculation

$$\Delta x_0 = x_1 - x_0 = 157.8 - 102.0 = 55.8\,mi$$

First speed interval calculation

$$|\bar{v}|_0 = \frac{\Delta x_0}{\Delta t_0} = \frac{55.8\,mi}{0.99\,hr} = 56.55\,mph$$

The overall average speed is

$$|\bar{v}| = \frac{\Sigma|\bar{v}|_i}{n} = \frac{56.55+57.35+48.82+63.74+61.19+56.63}{6} = 57.38\,mph$$

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Question 2

Given:

Time t, seconds	0	1	2	3	4	5	6	7	8
Position x, meters	0	4.1	7.53	10.92	12.31	12.35	11.83	10.49	7.95

Find:
calculate the velocity and acceleration of a moving ball at t = 2, 3, 4, 5, and 6 seconds using

a. forward finite difference
b. backward finite difference
c. centered finite difference

Diagram:

Theory:

Forward finite difference $\Delta x_i = x_{(i+1)} - x_i$

Backward finite difference $\Delta x_i = x_i - x_{(i-1)}$

Centered finite difference $\Delta x_i = x_{(i+1)} - x_{(i-1)}$

Velocity $v = \frac{dx}{dt}$

Acceleration $a = \frac{dv}{dt}$

Assumptions:
Our data is sufficiently smooth and differentiable across the given domain, as well as doubly differentiable for acceleration.

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Solution:

Time t, seconds	0	1	2	3	4	5	6	7	8
Position x, meters	0	4.1	7.53	10.92	12.31	12.35	11.83	10.49	7.95
Velocity v, m/s (forward)		3.43	3.39	1.39	0.04	-0.52	-1.34	-2.54	
Velocity v, m/s (backward)		4.1	3.43	3.39	1.39	0.04	-0.52	-1.34	
Velocity v, m/s (centered)		3.765	3.41	2.39	0.715	-0.24	-0.93	-1.94	
Acceleration a, m/s^2 (forward)			-2	-1.35	-0.56	-0.82	-1.2		
Acceleration a, m/s^2 (backward)			-0.67	-0.04	-2	-1.35	-0.56		
Acceleration a, m/s^2 (centered)			-0.68	-1.347	-1.32	-0.82	-0.85		

Specific velocity calculated where $t = 3$	
Forward finite difference velocity	$v_3 = \frac{x_{3+1} - x_3}{(3+1) - 3} = \frac{12.31 - 10.92}{1} = 1.39\ \frac{m}{s}$
Backward finite difference velocity	$v_3 = \frac{x_3 - x_{3-1}}{3 - (3-1)} = \frac{10.92 - 7.53}{1} = 3.39\ \frac{m}{s}$
Centered finite difference velocity	$v_3 = \frac{x_{3+1} - x_{3-1}}{(3+1) - (3-1)} = \frac{12.31 - 7.53}{2} = 2.39\ \frac{m}{s}$
Specific acceleration calculated where $t = 5$	
Forward finite difference acceleration	$a_5 = \frac{v_{5+1} - v_5}{(5+1) - 5} = \frac{-0.52 - 0.04}{1} = -0.82\ \frac{m}{s^2}$
Backward finite difference acceleration	$a_5 = \frac{v_5 - v_{5-1}}{5 - (5-1)} = \frac{-0.024 - 0.715}{1} = -1.35\ \frac{m}{s^2}$
Centered finite difference acceleration	$a_5 = \frac{v_{5+1} - v_{5-1}}{(5+1) - (5-1)} = \frac{-0.93 - 0.715}{2} = -0.82\ \frac{m}{s^2}$

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Question 3

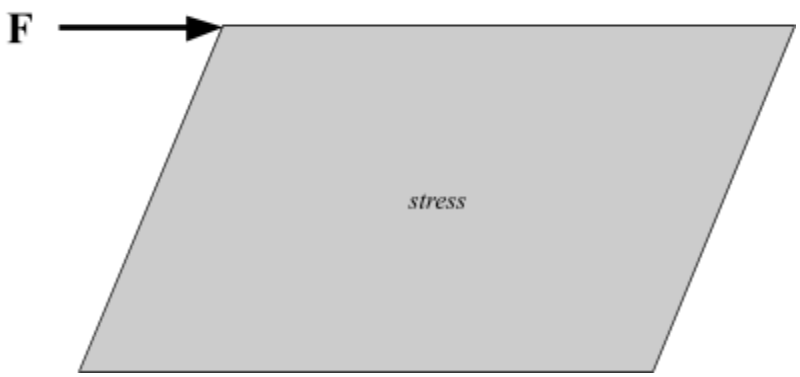
Given:

Dynamic viscosity $u = 1.8 \cdot 10^{-5}$.

y (m)	0	0.002	0.006	0.012	0.018	0.024
u (m/s)	0	0.067	0.572	2.291	5.047	9.041

Find: Sheer stress τ.

Diagram:



Theory:

Forward finite difference $\Delta x_i = x_{(i+1)} - x_i$

Backward finite difference $\Delta x_i = x_i - x_{(i-1)}$

Centered finite difference $\Delta x_i = x_{(i+1)} - x_{(i-1)}$

Shear stress $\tau = \mu \frac{du}{dy}$

Assumptions:

Our data is sufficiently smooth and differentiable across the given domain.

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Solution:

(index)	0	1	2	3	4	5
y (m)	0	0.002	0.006	0.012	0.018	0.024
u (m/s)	0	0.067	0.572	2.291	5.047	9.041
τ (N/m^2)		1.72e-03	4.00e-03	6.71e-03	1.01e-02	

Shear stress at *y = 0.012 (index 3)*

$$\tau_3 = \mu \frac{du}{dy}\Big|_3 = \mu \frac{u_{3+1} - u_{3-1}}{y_{3+1} - y_{3-1}} = \mu \frac{u_4 - u_2}{y_4 - y_2} = \mu \frac{5.047 - 0.572}{0.018 - 0.006} = \mu(372.92) = 6.71 \cdot 10^{-3}$$

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Question 4

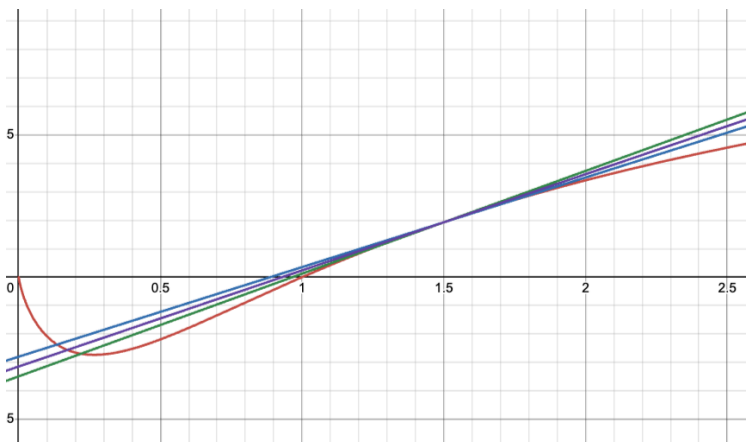
Given:

$f(x) = \frac{ln(x)sinh(x)}{e^x} \quad \Delta x = 0.25$

$f'(1.5) = 0.336925$

Find:
Estimate the derivative numerically using forward, backward, and centered finite differences, and determine the percent error between the true value and each of the estimated values.

Diagram:



Theory:

Forward finite difference $\Delta x_i = x_{(i+\Delta x)} - x_i$

Backward finite difference $\nabla x_i = x_i - x_{(i-\Delta x)}$

Centered finite difference $\delta x_i = x_{(i+\Delta x)} - x_{(i-\Delta x)}$

Percent error $\epsilon = |\frac{true-estimated}{true}| \cdot 100\%$

Assumptions:
Our function is differentiable across the given domain.

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Solution:

Measurements where $t = 1.5$ and $\Delta x = 0.25$		
Label	**Estimation**	**Percent error**
Forward finite difference	$\Delta f'(x) = \frac{f(x+\Delta x)-f(x)}{\Delta x}$ $= \frac{f(1.5+0.25)-f(1.5)}{0.25} = 0.314877$	$\epsilon = \lvert\frac{0.336925-0.314877}{0.336925}\rvert \cdot 100\%$ $= 6.544\%$
Backward finite difference	$\nabla f'(x) = \frac{f(x)-f(x-\Delta x)}{\Delta x}$ $= \frac{f(1.5)-f(1.5-0.25)}{0.25} = 0.360908$	$\epsilon = \lvert\frac{0.336925-0.360908}{0.336925}\rvert \cdot 100\%$ $= 7.118\%$
Centered finite difference	$\delta f'(x) = \frac{f(x+\Delta x)-f(x-\Delta x)}{2\cdot\Delta x}$ $= \frac{f(1.5+0.25)-f(1.5-0.25)}{2\cdot 0.25} = 0.337890$	$\epsilon = \lvert\frac{0.336925-0.337890}{0.336925}\rvert \cdot 100\%$ $= 0.286\%$

Calculating the Δx needed to have a sufficiently small error.

$$\Delta x_{new} = \frac{0.286\%}{7.118\%}\Delta x_{old} = 0.04018(0.25) = 0.01005$$